

Phishing Email Analysis Report

Cyber Security Internship — Task 2 | ElevateLabs

Objective

Analyze a suspicious phishing email sample to identify phishing characteristics, social engineering tactics, and threat indicators used by attackers.

Email Sample Overview

Field	Value
From	security-alert@paypa1-support.com
To	user@gmail.com
Subject	URGENT: Your Account Has Been Suspended
Date	Mon, 30 May 2026 03:22:11 +0000
Reply-To	noreply@paypa1-support.com
Originating IP	192.168.45.23
SPF Check	FAIL
DKIM Check	None / Missing
DMARC Check	FAIL

Phishing Indicators Found

1. Spoofed Sender Email Address (Typosquatting)

The attacker used paypa1-support.com (number 1 instead of letter l) to impersonate PayPal. Real PayPal emails come from @paypal.com only. This is classic typosquatting.

2. Malicious / Mismatched URL

The link shown leads to: <http://paypa1-secure-login.xyz/verify?token=abc123> Issues: Uses HTTP (not HTTPS), domain ends in .xyz (not paypal.com), contains fake paypa1 spelling, and has suspicious query parameters.

3. Urgent and Threatening Language

The email uses fear tactics: 'SUSPENDED', 'PERMANENTLY CLOSED within 24 hours', 'legal action may be initiated', 'all funds will be frozen'. This is social engineering to force the victim to act without thinking.

4. Requesting Sensitive Personal Information

The email asks for Credit Card Number, CVV, Expiry Date, Social Security Number, and Date of Birth. Legitimate companies NEVER request this via email.

5. Spelling and Grammar Errors

Multiple errors found: 'messsage' (message), 'Pleese' (Please), 'immedaitely' (immediately), 'reservd' (reserved). Common in phishing emails sent by non-native speakers or automated tools.

6. Email Header Discrepancies (SPF/DKIM/DMARC Failed)

All three email authentication checks FAILED: SPF (sender not authorized), DKIM (no digital signature), DMARC (domain policy violated). This confirms the email did NOT originate from real PayPal servers.

7. Sent at Unusual Time

Email was sent at 03:22 AM. Attackers often send phishing emails at odd hours when users are less vigilant or security monitoring may be reduced.

8. Generic Greeting

Uses 'Dear Valued Customer' instead of the user's real name. Legitimate companies always address you by your registered name. Generic greetings indicate mass phishing campaigns.

Phishing Indicators Summary

#	Indicator	Severity
1	Spoofed sender domain (paypa1)	HIGH
2	Malicious URL with fake domain	HIGH
3	Urgent / threatening language	HIGH
4	Requesting sensitive information	HIGH
5	Spelling and grammar errors	MEDIUM
6	SPF / DKIM / DMARC all FAILED	HIGH
7	Sent at unusual time (3 AM)	MEDIUM
8	Generic greeting used	MEDIUM

VERDICT: CONFIRMED PHISHING EMAIL

Tools Used for Analysis

Tool	Purpose
MxToolbox Header Analyzer	Analyze raw email headers
Google Admin Toolbox	Verify SPF / DKIM / DMARC records
VirusTotal	Scan suspicious URLs for malware
WhoisXML	Check domain registration info

How to Stay Protected

- Never click links in unsolicited emails — go directly to the official website
- Always check the sender email domain carefully for typos or misspellings
- Hover over links before clicking to reveal the real destination URL
- Enable Two-Factor Authentication (2FA) on all important accounts

- Report phishing emails to your IT/security team immediately
- Never share sensitive information (passwords, card numbers) via email
- Use antivirus software with real-time email scanning capability
- Analyze email headers using MxToolbox or Google Admin Toolbox

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